INCH-POUND

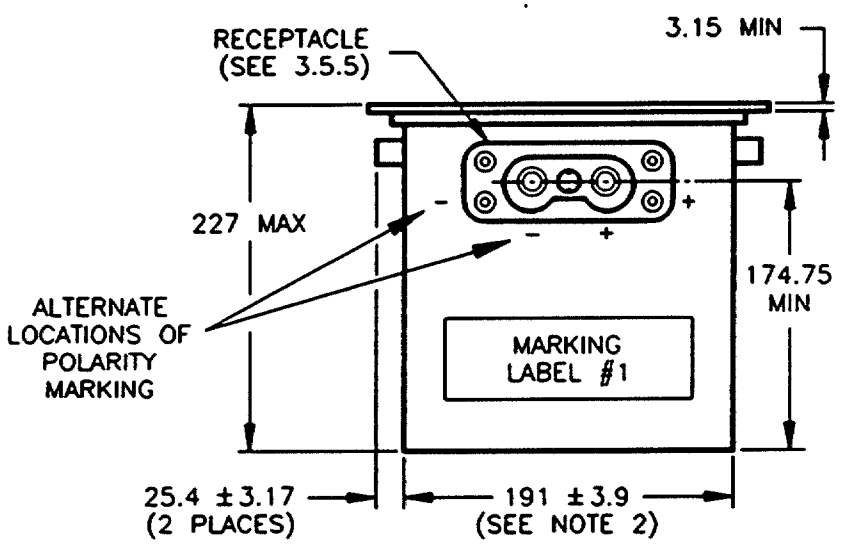
MIL-B-8565/9(AS)
29 October 1993

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 24-AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-8565



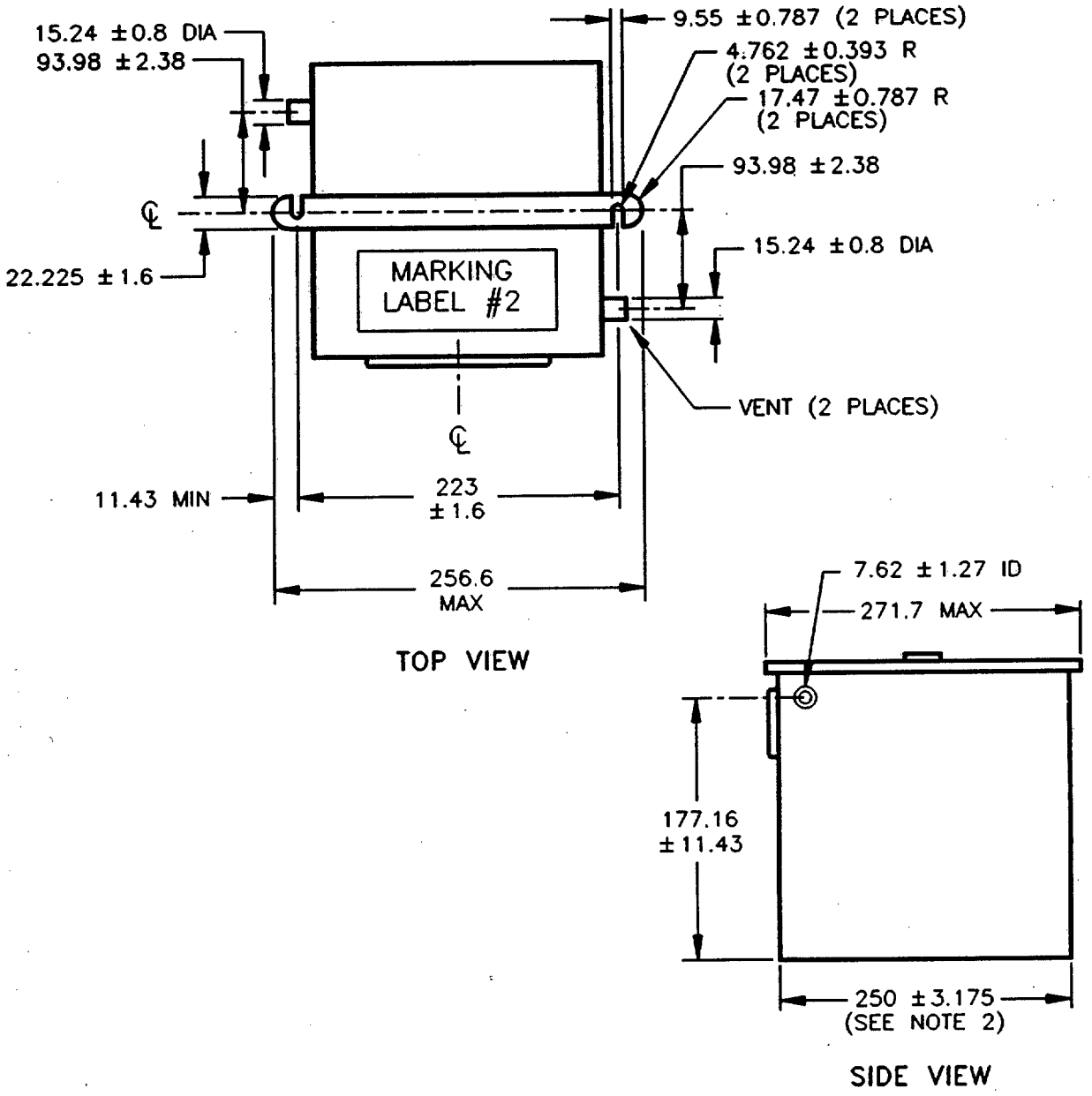
FRONT VIEW

NOTES:
1. All dimensions are in millimeters.
2. Allow additional 12 mm width for cover.
3. Abbreviations used in this figure are maximum (MAX), minimum (MIN) and part number (PN).

FIGURE 1. PN D8565/9-1 dimensions.

AMSC N/A FSC 6140
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

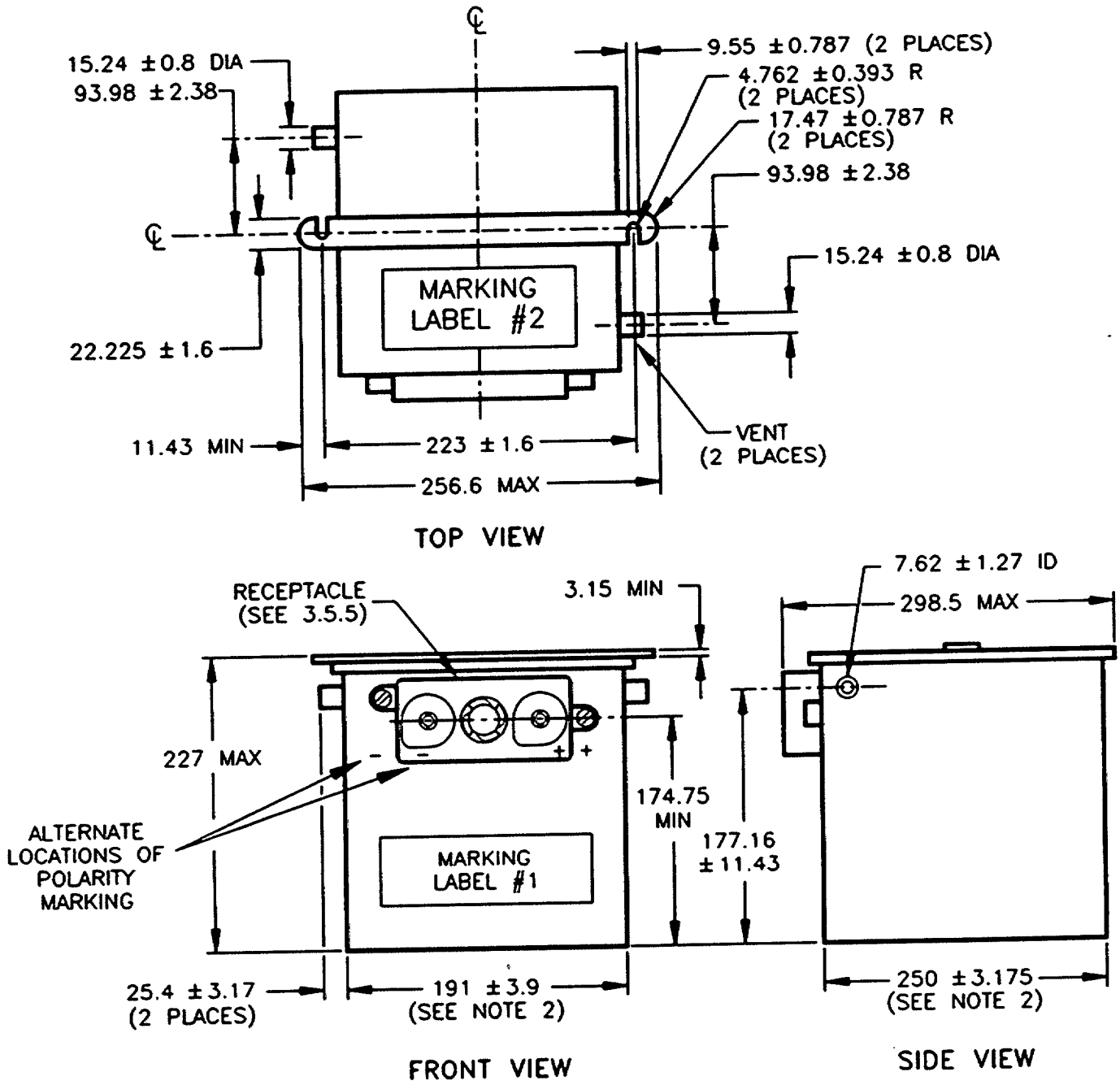
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NOTES:
1. All dimensions are in millimeters.
2. Allow additional 12 mm length for cover.
3. Abbreviations used in this figure are diameter (DIA), inside diameter (ID), maximum (MAX), minimum (MIN), part number (PN) and radius (R).

FIGURE 1. PN D8565/9-1 dimensions-Continued.

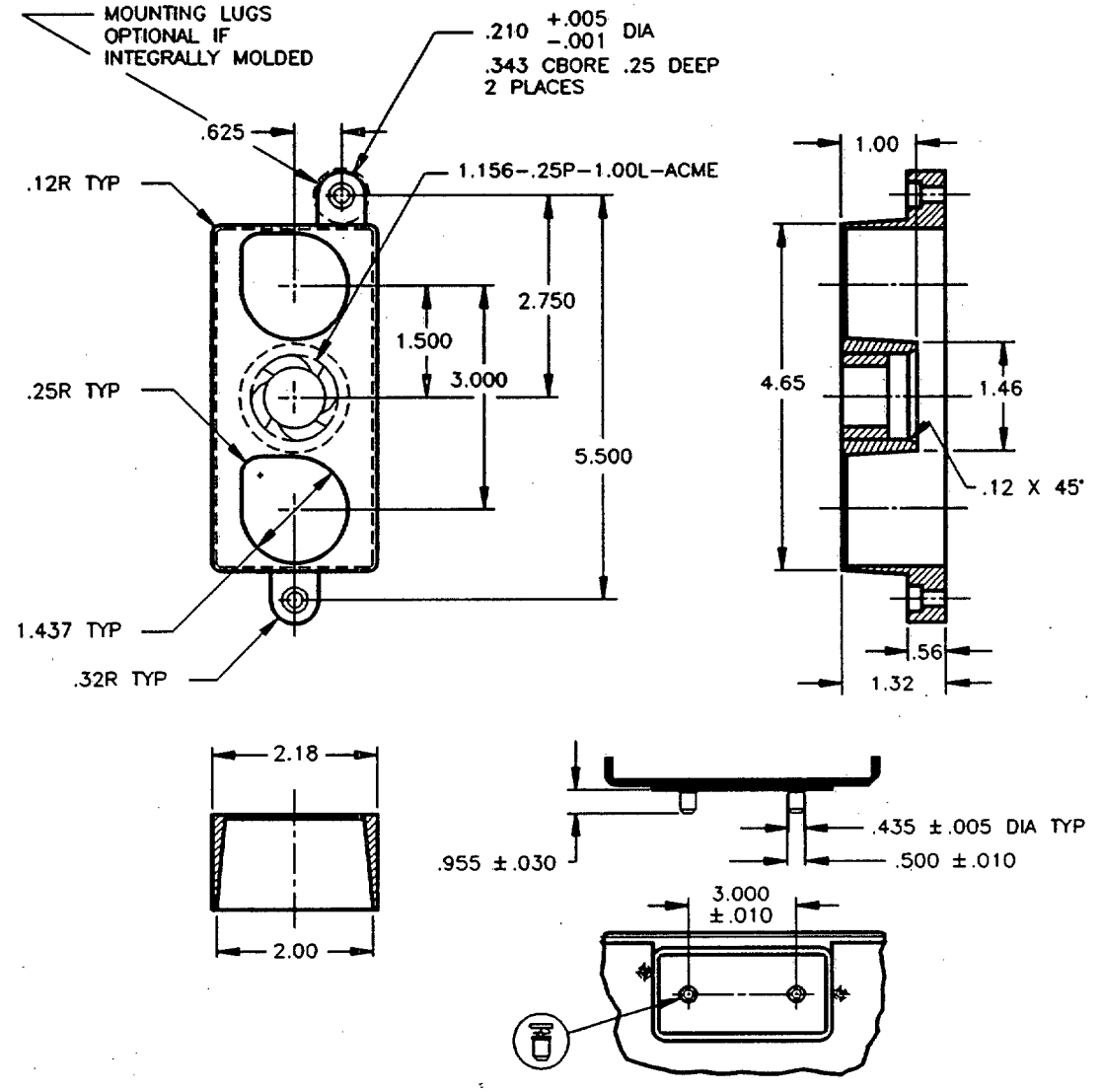
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NOTES:
1. All dimensions are in millimeters.
2. Allow additional 12 mm width and length of cover.
3. Abbreviations used in this figure are diameter (DIA), inside diameter (ID), maximum (MAX), minimum (MIN), part number (PN) and radius (R).

FIGURE 2. PN D8565/9-2 dimensions.

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NOTES:

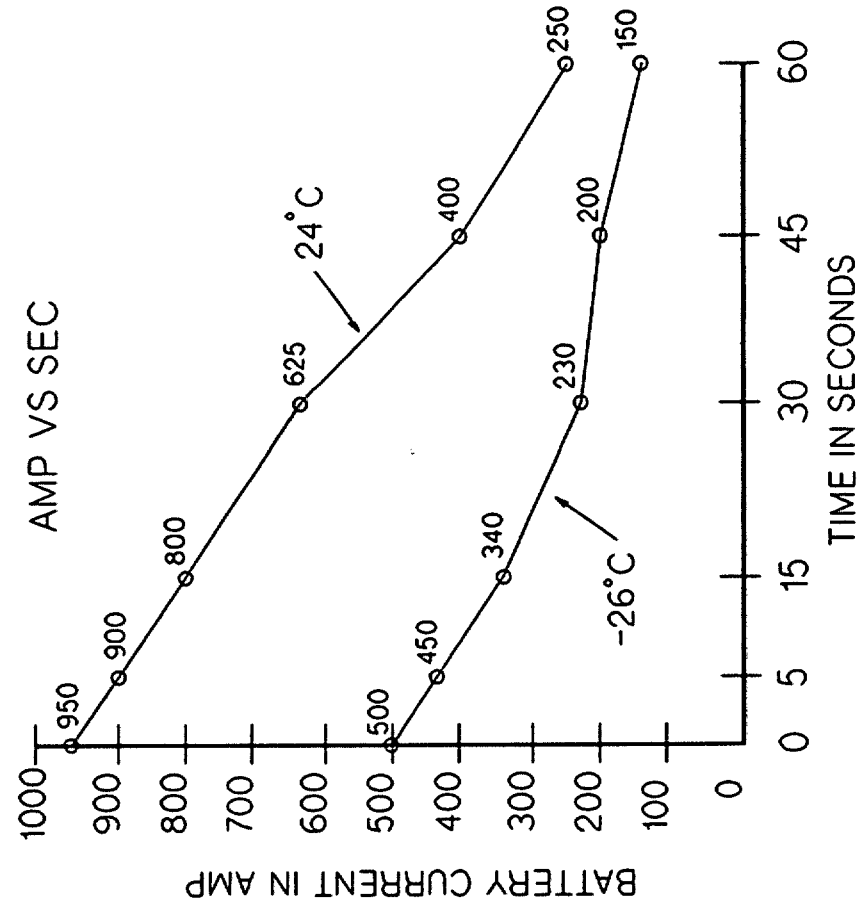
1. Material and finish to be acid resistant.
2. All dimensions are in inches.
3. Tolerances are X.XX (± .03 inches), X.XXX (± .010 inches).
4. Receptacle to case attachment method optional for nonmetallic containers.
5. Draft angles allowed in conformance with commercial die casting practice.
6. Abbreviations used in this figure are counterbore (CBORE), diameter (DIA) and typical (TYP).

FIGURE 3. Typical receptacle adapter.

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MINIMUM DISCHARGE CURRENT VS TIME		
TIME IN SECONDS	CURRENT IN AMP	
	AT 24°C	AT -26°C
0+	950	500
5	900	450
15	800	340
30	625	230
45	400	200
60	250	150

AVERAGE DISCHARGE CURRENT DURING 14-V DISCHARGE SHALL BE NOT LESS THAN THE FOLLOWING AMP VALUES	
24°C	610.0 AMP
-26°C	275.0 AMP



NOTE:

1. Abbreviations used in this figure are amperes (AMP), seconds (SEC), versus (VS), volts (V), and degree Celsius (°C).

FIGURE 4. Constant 14-V discharge curves.

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REQUIREMENTS:

1. The dimensions of the batteries shall be as shown on figure 1 and figure 2.

2. The part number of the battery shall be D8565/9-1 for figure 1 and D8565/9-2 for figure 2.

3. The rated capacity of the battery shall be 24 ampere-hours based on its ability to conform to the life test final discharge. The life test final discharge is a 1-hour discharge at 24°C ambient temperature with a 24-ampere load (1C-rate) to a cutoff voltage of 18 volts.

4. The weight of the battery shall be not greater than 28.6 kg.

5. Battery life in the aircraft shall be 3 years.

6. Table I lists the quality assurance tests.

7. The battery shall conform to MIL-B-8565 for type I, medium rate, except as modified by the following.

3.5.5 Receptacles. Add the following: "For D8565/9-1, the female receptacle (front view) shall be dimensionally equal to MS3509, style 1. For D8565/9-2, the female receptacle shall be dimensionally equal to figure 3 of the applicable specification sheet. Either receptacle may be integrally molded in the container."

3.6.8 Capacity and electrical performance. Add the following: "During the 1-minute, 14-volt, constant voltage discharge, the battery shall meet at least one of the following conditions:

"Condition 1. The discharge current during the constant voltage discharge shall be not less than the ampere values shown on figure 4 of the applicable specification sheet. Inability of the battery to deliver the specified minimum discharge currents during the discharge shall constitute a requirement failure.

"Condition 2. The discharge current during the constant voltage discharge shall be not less than the ampere values shown on figure 4 of the applicable specification sheet throughout the first 15 seconds of the discharge. Furthermore, through the duration of the discharge, the battery shall maintain an average current not less than the average ampere value shown on figure 4 of the applicable specification sheet. Inability of the battery to deliver the specified currents during the discharge shall constitute a requirement failure."

Supersession Data. D8565/9-1 supersedes M83769/4-1.
D8565/9-2 supersedes M83769/2-1.

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TABLE I. Inspection of batteries and order of test.

Order of test	Description of inspection	Requirements paragraph	Inspection method paragraph	Qualification inspection			Quality conformance inspection		
				Qualification sample number			Group A tests	Group B tests	
				1	2	3		1	2
1	Visual & mechanical examination	3.5, 3.6.1, 3.6.3 & 3.6.22	4.6.3	X	X	X	X	X	X
2	Dimensions & weight	3.6.5	4.6.4	X	X	X	X	X	
3	Dielectric strength	3.6.4	4.6.5	X	X	X	X	X	X
4	Strength of vent tubes	3.6.7	4.6.7	X	X	X	X	X	X
5	Color & marking	3.6.3	4.6.8	X	X	X	X	X	X
6	Conditioning charge	4.5.1.4	4.6.9	X	X	X		X	X
7	Capacity discharge	3.6.8	4.6.10	X	X	X	X	X	X
8	Engine start discharge curve	3.6.8	4.6.20.1	X	X		X	X	
9	Strength of receptacle	3.6.9	4.6.12			X			X
10	Life cycling	3.6.10	4.6.13	X				X	
11	Capacity discharge at 49°C	3.6.8	4.6.14		X				
12	Charge & discharge test at low temperature	3.6.8	4.6.15		X				X
13	Discharge while inverted	3.5.8 & 3.6.8	4.6.16		X				X
14	Altitude	3.6.11 & 3.6.12	4.6.17		X				X
15	Mechanical shock	3.6.11 & 3.6.12	4.6.18	X					
16	Temperature shock	3.6.11 & 3.6.12	4.6.19	X				X	
17	Temperature rise & float	3.6.12 & 3.6.13	4.6.21		X				X
18	Battery gas emission test	3.6.11 & 3.6.14	4.6.22		X				
19	Vibration	3.6.11, 3.6.12 & 3.6.16	4.6.23		X				
20	Humidity	3.6.11, 3.6.12 & 3.6.17	4.6.24		X				X
21	Salt fog	3.6.11, 3.6.12 & 3.6.18	4.6.25	X				X	
22	Ground storage	3.6.15.1	4.6.26			X			
23	Physical integrity at high temperature	3.6.11 & 3.6.12	4.6.27	X					
24	Impact resistance test	3.6.19	4.6.27.1		1/				
25	Final examination	3.6.4	4.6.5 & 4.6.30	X	X	X		X	X

1/ For nonmetallic battery container designs, two outer nonmetallic battery containers must be supplied. The containers shall be subjected to the impact resistance test of 4.6.27.

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